

PLANRS ACADEMY

Science Art Academy: Grade 3 Chemistry

The Secret Life of Ink

Chromatography Mystery

Session 1: The Great Colour Race

A Mysterious Note!

Ananya found a secret note in the classroom written with a black sketch pen. She thought black was just one colour, but what if she's wrong? What if black ink is hiding a whole team of colours inside? Today, we use science to uncover the truth!

Your Science Art Goals

Today, we will become "Art-Scientists" to achieve these goals:

-  **Understand Mixtures:** Learn that some things are a "team" of many parts.
-  **Define Chromatography:** Discover the method used to separate hidden parts.
-  **Perform the Experiment:** Watch the "Colour Race" happen on paper.
-  **Become a Science Artist:** Use chromatography to create beautiful patterns.



THE QUICK GUESS:

What colours do you think are hidden inside a black sketch pen? Red? Blue? Yellow? Let's make a prediction before we start!

What Is a Mixture?

In science, a **Mixture** is when two or more things are put together but not permanently joined. Ink is a perfect example of a hidden mixture.

The Secret Team:

Black ink looks like one colour, but it is actually many different colours hidden together like a team!



The Fruit Chaat Analogy

Think about a bowl of Mixed Fruit Chaat.

- ◆ It looks like one tasty snack.
- ◆ But inside, you can find pieces of Apple, Banana, and Papaya.
- ◆ Black ink is just like a "chaat" of different colours!

[Visual: Comparison diagram of Mixed Fruit Chaat and a drop of Black Ink]

The Great Colour Race

If ink is a mixture, how do we separate the colours? We use a magic trick called **Chromatography**.

Definition: Chromatography is a way to separate a mixture by making its parts move at different speeds through a material.



The Rainy Day Race

Imagine a race on a rainy day in your colony.

- 💧 When the ground gets wet, some children run very fast.
- 💧 Some children run a bit slower on the slippery ground.
- 💧 In chromatography, water helps the colours "run" up the paper.

Fast colours reach the top first, while slow ones stay behind!

The Experiment: Uncovering the Secret

Let's set up our own Colour Race. You will be amazed at what you see!

What You Need:

- ✓ A black sketch pen (water-soluble).
- ✓ A strip of tissue paper or filter paper.
- ✓ A small drop of water.
- ✓ A flat surface (newspaper or table).



ART-SCIENTIST TIP:

Different brands of black pens use different colour "teams." Try two different black pens to see if the results are the same!

Step-by-Step Magic

1 The Starting Line: Draw a thick black line or a big dot near the bottom of your paper strip.

2 Add the "Runner": Place a small drop of water directly on or just below the black ink.

3 Watch the Race: Wait for a few minutes. Watch as the water starts to climb up the paper, carrying the ink with it!

[Visual: Illustration of water carrying different colours up a tissue paper strip]

What Do You See?

As the water climbs, the "Black" ink will start to disappear. In its place, you might see:

-  **Bright Blue**
-  **Deep Purple**
-  **Warm Brown**
-  **Soft Orange**



The Science Secret

The colours that you see at the top of the paper are the "**Fast Runners.**"

The colours that stay near the bottom are the "**Slow Runners.**"

Interactive: Hidden Colours Challenge

Can you guess which colours are hidden in these other sketch pens?

1. THE GREEN PEN

Which two primary colours do you think are hiding in Green?

Your Guess: _____ and _____

2. THE PURPLE PEN

Which two colours make up the team for Purple?

Your Guess: _____ and _____

3. THE ORANGE PEN

Which team is hiding here?

Your Guess: _____ and _____

Be a Science Artist!

Chromatography is not just for scientists — it's for artists too!

The Colour Map Worksheet

Try chromatography with 3 different pens. In the boxes below, draw the "Colour Map" you see after the water finishes the race.

[Pen 1]

[Pen 2]

[Pen 3]

🌟 *Art Idea: Use these colourful strips to make a bookmark or a greeting card!* 🌟

Lesson Recap

You have successfully uncovered the secret life of ink! Let's remember our key findings:

✨ **Mixtures:** Black ink is actually a "team" of many hidden colours.

✨ **Chromatography:** We can separate these colours using water.

✨ **The Race:** Different colours move at different speeds on paper.

✨ **Art-Scientist:** Science can help us create beautiful things!

STAY CURIOUS!

Next time you use a sketch pen, remember the hidden team of colours inside!